Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 2 – Further Differentiation

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Differentiate $y = (4x^2 - 3x + 1)^4 \ln(3x - 2)$. **[6]**

2. Differentiate $y = x^2 \ln(2x + 1)$. **[4]**

3. Differentiate $y = e^{2x} \sin 3x$. **[4]**

4. Differentiate $y = \ln\left(\frac{2x+1}{x^2+3}\right)$. **[4]**

5. Differentiate $y = \frac{\ln x}{x}$. **[4]**

6. $x^2 + xy + y^2 = 7$. Find $\frac{dy}{dx}$. **[5]**

7. $x^2 + y^2 = 2xy + 3$. Find $\frac{dy}{dx}$. **[5]**

8. $ye^x = x + y$. Find $\frac{dy}{dx}$. **[5]**

9. Parametric: $x = 3t^2 - 2$, $y = t^3 - 4t$. Find $\frac{dy}{dx}$ in terms of t. **[5]**

10. Parametric: $x = 2t + 1$, $y = t^2 - 3$. Find $\frac{dy}{dx}$ at $t = 2$. **[5]**

11. Parametric: $x = \sin t$, $y = \cos 2t$. Find $\frac{dy}{dx}$. **[5]**

12. Differentiate $y = \arcsin(2x)$. **[3]**

13. Differentiate $y = \arccos(x^2)$. **[4]**

14. Differentiate $y = \arctan(3x)$. **[3]**

15. Differentiate $y = \ln(\sin x)$. **[3]**

Section B – Hard (Q16–25)

16. Parametric $x = 3t^2 - 2$, $y = t^3 - 4t$. Equation of tangent at $t = 2$. **[7]**

17. $y = \arcsin(2x)$. Equation of normal at $x = \frac{1}{4}$. **[6]**

18. $ye^{2x} = x^2 + 1$. Find $\frac{dy}{dx}$ in terms of x, y. **[6]**

19. $x^3 + y^3 = 3xy$. Find $\frac{dy}{dx}$ at $(1, 2)$. **[6]**

20. Parametric $x = t^2$, $y = t^3 - 3t$. Find coordinates of stationary points. **[7]**

21. Differentiate $y = x^x$ (write as $e^{x \ln x}$). **[5]**

22. Find the equation of the tangent to $y = \ln(x^2 + 1)$ at $x = 1$. **[5]**

23. $x = \cos t$, $y = \sin t$. Show $\frac{dy}{dx} = -\cot t$. **[4]**

24. Find $\frac{d^2y}{dx^2}$ if $y = \ln(1 + x^2)$. **[5]**

25. The curve $xy = e^{x+y}$. Find $\frac{dy}{dx}$ in terms of x and y. **[6]**

End of Worksheet • *Show all working clearly.*